OPERATING SYSTEMS

UNIT IV – MEMORY MANAGEMENT STRATEGIES & MASS STORAGE

SECTION A – 2-MARK QUESTIONS (3-MARK ANSWER FORMAT)

Answers, examples and simple diagrams

Q1. Define Dynamic Linking.

Definition:

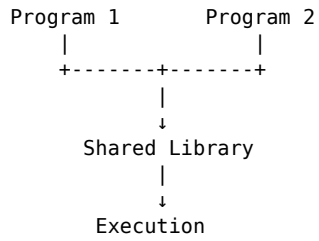
Dynamic Linking is a method of connecting library files to a program during execution time instead of loading them before execution.

Explanation:

- Library functions are connected when the program runs.
- It reduces memory usage.
- Multiple programs can share the same library.

Example: Many programs can use the same shared library without keeping separate copies.

Diagram:



Q2. What is a Page Fault?

Definition:

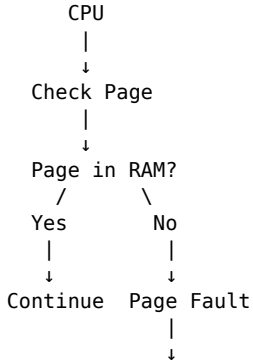
A Page Fault occurs when a program tries to access a page that is not currently available in main memory (RAM).

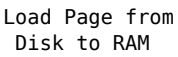
Explanation:

- The CPU checks whether the required page is in RAM.
- If the page is not available, a page fault occurs.
- The OS brings the required page from secondary storage into RAM.

Example: When we open an application and its required data is not in RAM, the OS loads it from storage.

Diagram:





Q3. What is Paging in Memory Management?

Definition:

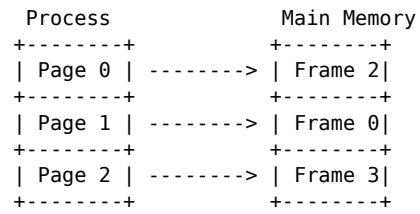
Paging is a memory management technique that divides a process into fixed-size blocks called Pages and physical memory into fixed-size blocks called Frames.

Explanation:

- A process is divided into pages.
- RAM is divided into frames.
- Pages are loaded into available frames.

Example: A book is divided into pages, and those pages can be kept in different places.

Diagram:



Q4. Define Demand Paging.

Definition:

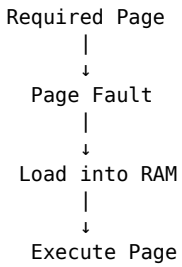
Demand Paging is a memory management technique in which a page is loaded into RAM only when it is needed.

Explanation:

- All pages are not loaded at the beginning.
- When a required page is missing, a page fault occurs.
- The OS loads that page into RAM.

Example: We download a file only when we need it instead of downloading everything in advance.

Diagram:



Q5. What is Thrashing in Operating Systems?

Definition:

Thrashing is a condition in which the system spends more time moving pages between RAM and disk instead of executing programs.

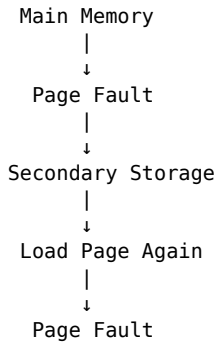
Explanation:

- Too many page faults occur.
- The OS continuously moves pages between RAM and disk.

- System performance becomes very slow.

Example: A computer becomes very slow because it continuously loads and removes pages from memory.

Diagram:



Q6. Define Swapping in Memory Management.

Definition:

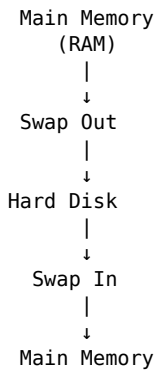
Swapping is a memory management technique in which a process is temporarily moved from RAM to secondary storage and brought back when needed.

Explanation:

- A process is moved from RAM to disk.
- This creates free space in RAM.
- The process is brought back into RAM when required.

Example: Moving an unused application out of memory to make space for another application.

Diagram:



Q7. Differentiate between Internal Fragmentation and External Fragmentation.

Definition:

Fragmentation means wastage of memory space during memory allocation.

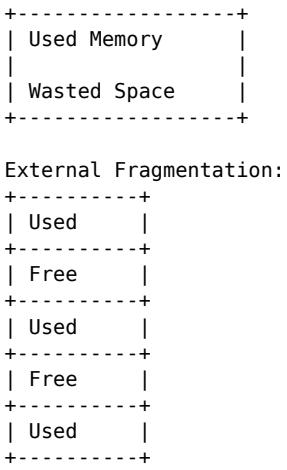
Differences:

Internal Fragmentation: Wasted space inside an allocated memory block; happens when allocated space is larger than required. External Fragmentation: Free memory is available in small separate blocks; happens when free memory is divided into small pieces.

Example: Internal Fragmentation: Fixed-size partitioning. External Fragmentation: Variable-size partitioning.

Diagram:





Q8. What is Translation Lookaside Buffer (TLB)?

Definition:

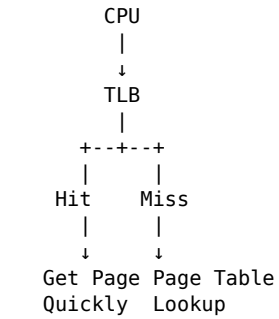
TLB is a small and fast memory used to store recently used page table entries.

Explanation:

- It stores recently used page-to-frame mappings.
- It helps the CPU find the required page quickly.
- It reduces memory access time.

Example: TLB is like a shortcut list that helps us find information quickly instead of searching a big book.

Diagram:



Q9. What is Copy-on-Write?

Definition:

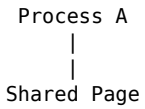
Copy-on-Write (COW) is a technique in which two processes share the same memory until one process tries to modify it.

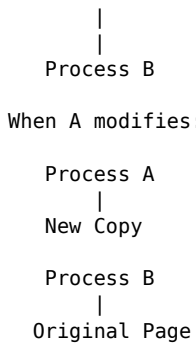
Explanation:

- Processes initially share the same memory pages.
- When one process modifies a page, a separate copy is created.
- It saves memory and improves efficiency.

Example: Two students use the same book. If one student wants to write in it, a separate copy is given.

Diagram:





Q10. What is RAID? State its main purpose.

Definition:

RAID stands for Redundant Array of Independent Disks. It is a technique that combines multiple hard disks into one logical storage system.

Explanation:

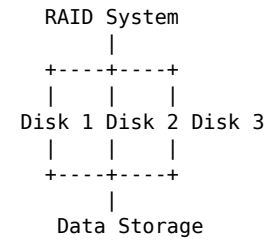
- It combines multiple disks.
- It improves data storage performance.
- It provides data protection against disk failure in supported RAID levels.

Main Purpose:

Improve storage performance. Increase reliability. Protect data using redundancy.

Example: RAID is used in servers to store important data safely.

Diagram:



Unit IV – Quick Revision

Q.No	Topic	Important Keywords
1	Dynamic Linking	Library, Execution Time
2	Page Fault	Page Missing, RAM
3	Paging	Pages, Frames
4	Demand Paging	Load When Needed
5	Thrashing	Too Many Page Faults
6	Swapping	RAM, Disk, Swap In/Out
7	Fragmentation	Internal, External
8	TLB	Fast Memory, Page Table
9	Copy-on-Write	Shared Page, Separate Copy
10	RAID	Multiple Disks, Reliability